

MGTA 444 Business Analytics Consulting and Solution Design

Winter 2026

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DESCRIPTION

Welcome to our graduate course on Business Analytics Solution Design! In this class, we'll dive into the art and science of creating effective business analytics solutions, recognizing that designing the right solution is both the most critical and the most challenging part of any project. Albert Einstein's famously said, "If I had an hour to solve a problem, I'd spend 55 minutes thinking about the problem and 5 minutes thinking about solutions." We'll also stress the importance of thoroughly understanding and defining problems before jumping into solutions. You'll learn how to frame problems concisely, work efficiently through the data analysis project flow, and apply proven strategies to avoid common mistakes. Through case studies and hands-on assignments, you'll gain skills needed to craft data-driven solutions that make a difference.

OBJECTIVES

At the end of this course, you will:

- Strengthen your ability to define and frame business problems accurately before designing solutions.

- Master techniques for translating business challenges into solvable analytical problems.
- Apply proven principles and methodologies to avoid common pitfalls in solution design.
- Gain practical experience through case studies and projects to create data-driven solutions that align with strategic business goals.
- Learn about model design and implementation to ensure comprehensive problem-solving approaches.

MATERIALS

- CLASS READER POSTED IN CANVAS
- REQUIRED TEXTBOOK AVAILABLE FROM LIBRARY: Instructions in Canvas.

Textbook title	Author	ISBN	Copyright Year	Edition	Edition Note	Publisher
Data Science for Business: What you need to know about Data Mining and Data Analytic Thinking	Tom Fawcett/Foster Provost	978-1449361327	2013	N/A	N/A	O'Reilly Media

SCHEDULE

Session	Time	Class Topic	Reading and Assignments for Next Class
0		Pre-class Reading	Read: Data Science for Business (Chapters 1, 2, and Appendix A: Be Ready to Evaluate Proposals for Data Science Projects) Read: Class Reader (Framing Data Science Problems the Right Way from the Start)
1	11:00-1:50 Break 12:00 Break 1:00	Problem Framing and Solution Design • Class introduction • Translating business needs into solvable analytical problems • Systematically approaching the analytical problem	Read: Data Science for Business (Chapter 3 and Appendix A: Proposal Review Guide) Read: Class Reader (Why Big Data and Data Science Problems Fail)
2	11:00-1:50 Break 12:00 Break 1:00	Data and Data Exploration • Data asset inventory • Dirty data • Insights and patterns ➤ Capstone Team Presentations	Read: Data Science for Business (Chapter 5) Read: Class Reader (Avoid ML Failures by Asking the Right Questions) Assignment: 1-page review of a capstone presentation (I)
3	11:00-1:50 Break 12:00 Break 1:00	Modeling Strategy and Solution Paths (1 of 2) • Modeling process flow • Model evaluation ➤ Capstone Team Presentations	Read: Data Science for Business (Chapter 11) Read: Class Reader (Avoid ML Failures by Asking the Right Questions) Assignment: 1-page review of a capstone presentation (I)
4	11:00-1:50 Break 12:00 Break 1:00	Prescriptive Analytics • ROI, Estimated Value, and Frameworks • Action Planning ➤ Capstone Team Presentations	Read: Data Science for Business (Chapter 5) Read: Class Reader (Avoid ML Failures by Asking the Right Questions) Assignment: 1-page review of a capstone presentation (I)
5	11:00-1:50 Break 12:00 Break 1:00	Deployment and Management of Solutions • Successful implementations Common process flows ➤ Capstone Team Presentations	Read: Data Science for Business (Chapter 5) Read: Class Reader (Avoid ML Failures by Asking the Right Questions) Assignment: Capstone Project Presentation (G)

- Class attendance is required.

ASSIGNMENTS

Assignments are assigned at the end of each class and due before the next class.

I – Independent, individual work only. No collaboration or consultation allowed.

G – Students may work together in groups and turn in one project or assignment for the entire group.

C – Collaboration with classmates is allowed. However, each student must submit an assignment.

EXAMS

No Final Exam

ATTENDANCE

Attendance is mandatory. One absence is allowed with prior approval from your TA. An unexcused absence is a one-half letter grade penalty for the course.

GRADING

Assignments	Points
Assignment 1	15 points
Assignment 2	20 points
Assignment 3	20 points
Assignment 4	20 points
Attendance	25 points (5 points per class)
	100 points possible

COURSE POLICIES

- The class is designed to be interactive, and active participation is strongly encouraged.
- AI tools may be used as supports in this class. However, submitted assignments must reflect your own work. Points will be deducted if submissions mirror the results generated from ChatGPT or other tools.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University

expects that both faculty and students will honor this principle and in so doing protect the validity of university intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind. The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires accommodation to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member. The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member to initiate the request for accommodation in classes, examinations, or other academic program activities. No accommodation can be implemented retroactively. Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.